

# GATE

## Governed, Assisted, Traceable Engineering

**An AI-assisted SDLC framework.** A software development life cycle that defines the stages, their inputs and outputs, the human control points and the actors of each one.

Technical reference document, agnostic of company, domain, language and platform.

It covers the whole journey: from someone raising a need to the system running in production, with the organisation having learned something from doing it.

Assistance that stops where judgement begins.

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## 1. The thesis

Most teams already use AI assistants. Almost always the same way: each person with their own prompt, in their own window, leaving no trace. The result shows within a few months.

Quality depends on who used the assistant that day and how well they asked. Decisions evaporate, and the same debate comes back three weeks later in another change. Security and compliance arrive once the code is written, which is the most expensive moment. And nothing is auditable: there is no way to answer which standard was applied to a change, who approved it, or since when it has been in force.

GATE adds no new capability to the assistant. It orders the ones it already has into stages with an input, an output and an owner, with explicit points where a person decides, and with mechanisms that turn every decision and every incident into a reusable standard.

The gain does not come from the assistant doing more. It comes from what it does sticking, chaining into the next step, and not depending on someone remembering.

### What kind of framework this is, and what it does not replace

GATE is an **SDLC**: it describes the life cycle of a software product, from the need to retirement, with the stages, artefacts and controls of each leg. It sits in the same family as the classic life cycle models and as continuous integration and delivery practice.

What sets it apart is that **AI assistance is in the design of the cycle, not layered on top of it**. In existing models the assistant is added as a tool each person uses at their own discretion. Here every stage declares what the machine produces, what the person decides, and what gets written down for the next stage.

**It does not replace how you organise the team.** Scrum, Kanban or whatever you use to prioritise, plan and coordinate keeps working on top of GATE. This framework defines no ceremonies, no cadence, no estimation, and no backlog shaping. It defines the journey of a piece of work and who clears each step.

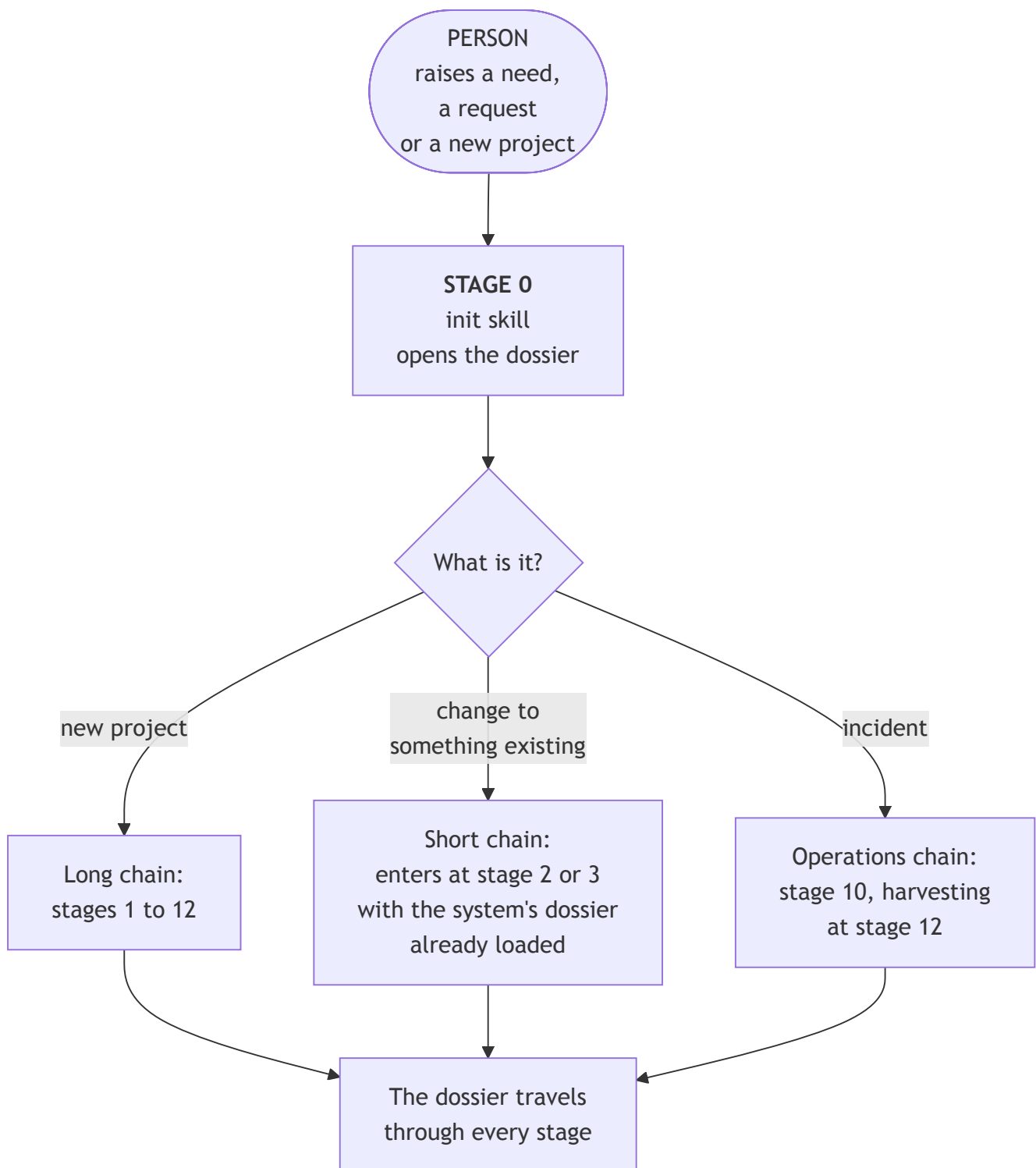
**It does not replace your engineering practices either.** Continuous integration, automated testing, infrastructure as code and progressive delivery are inputs to stages 5 through 9, not alternatives to them.

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## 2. Stage 0: the single entry point

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GATE has **one door**. A person raises a need, a request or a new project, and everything else chains from there.

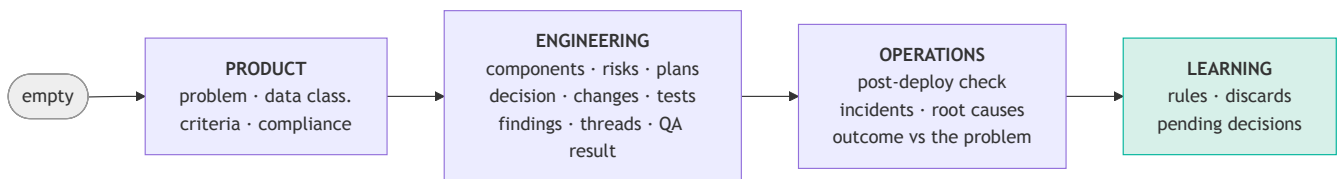


What the person hands over at stage 0 can be as informal as a transcribed voice note or a forwarded email. What **cannot** be missing is who is asking and what for. The first stage gathers everything else by asking. Each of those three chains is walked step by step in [START-HERE.md](#), with what it changes compared with working without the framework, and what it costs.

## The dossier

Stage 0 produces a **dossier**: a single artefact that travels with the work through the whole cycle, accumulating what each stage produces. It is what lets the stages chain without anyone retyping the context.

| Stage that writes it | What it adds to the dossier                                                |
|----------------------|----------------------------------------------------------------------------|
| 0 · Entry            | Who is asking, what for, and what type of work it is                       |
| 1 · Definition       | Problem, context, constraints, data classification                         |
| 2 · Requirements     | Functional and non-functional acceptance criteria, compliance requirements |
| 3 · Refinement       | Affected components, risks with owners, test and observability plans       |
| 4 · Design           | Architecture decision, alternatives ruled out and why                      |
| 5 · Development      | Proposed changes, instrumentation, tests                                   |
| 6 · Pre-review       | Findings resolved, functional verification script                          |
| 7 · Change review    | Threads, decisions, published findings                                     |
| 8 · Verification     | Result per criterion                                                       |
| 9 · Deployment       | Post-deployment check against the stage 2 criteria                         |
| 10 · Operations      | Incidents, root causes, recovery drill results                             |
| 11 · Validation      | Outcome measured against the stage 1 problem                               |
| 12 · Learning        | Rules harvested, discards with their reason, pending decisions             |



Nothing on that line is rewritten. Each stage adds and none re-asks. That is the point: context stops being rebuilt at every handover.

The dossier is also the audit record. It answers what was asked for, which standard was applied, who approved each step, and on what evidence.

## What chains on its own and what does not

The chain advances by itself between stages. **It stops at every human gate**, without exception.

A cycle that runs end to end without intervention is not this framework: it is a code factory with no owner. The gates are the product, not the obstacle.

### 3. Principles

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They apply to every stage. An implementation that breaks one stops being GATE.

**The assistant prepares, the person decides.** The assistant does the heavy lifting: reading, correlating, drafting, checking. Every consequential decision belongs to a person, and is taken on something already written down.

**Nothing runs unattended into shared systems.** Publishing a comment, opening a change, writing in a ticket, touching an environment, silencing an alert. All of it is shown first and executed on explicit approval, one at a time. Approving one thing does not authorise the rest.

**Verify before asserting.** No claim about how the system works rests on the assistant's memory or on documentation that may be stale. It is checked against the code, the data or the environment. If it could not be checked, that gets said.

**Every rule shows where it came from.** A standard with no link to the discussion or incident that produced it is someone's opinion in the shape of a norm.

**Discards get recorded.** When someone rejects a finding, the reason is written down. Without that the assistant raises it again and the team learns to ignore it, including the findings that mattered.

**Missing dependencies get named.** If an access, a token or a data source is missing, say which one and carry on with what you can, marking what was left uncovered.

**Check before creating.** Before opening a ticket, a change or a comment, check whether it already exists. If it does, update it. Never a duplicate.

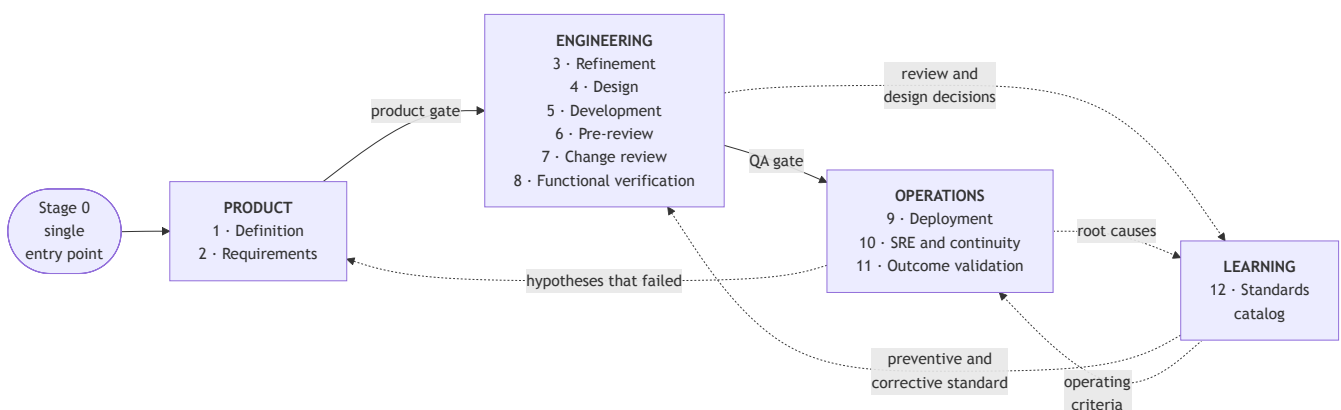
**Every stage delivers a verifiable artefact.** A stage whose output is "we agreed" did not happen.

**The dossier is the only source of context.** What is not in the dossier does not exist for the next stage. That forces the context to be written down instead of living in the head of whoever was in the meeting.

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### 4. The cycle

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The dotted arrows are the learning loop. A decision taken today in a review, or a root cause found today in an incident, becomes a standard that tomorrow applies on its own, before any code is written.

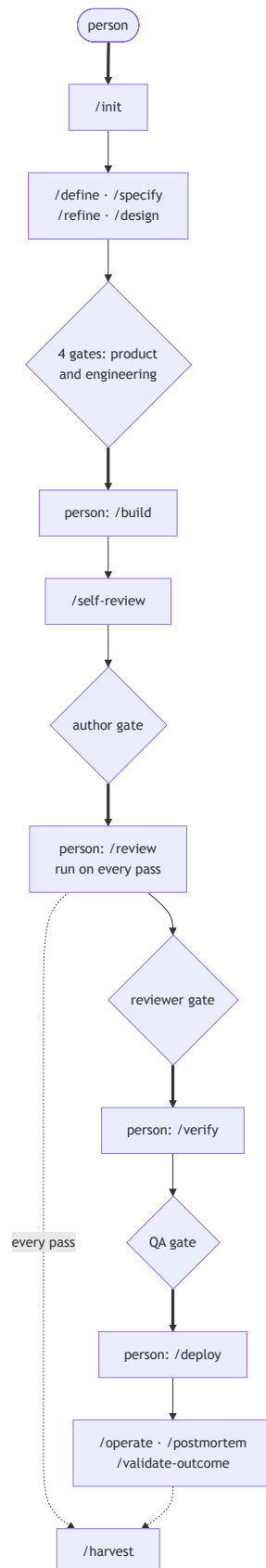
That the catalog also feeds refinement is what stops the standard from being merely corrective. A requirement that collides with a rule in force is caught before the first line.

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## **5. The skills, named**

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Fourteen cycle skills and two cross-cutting ones. The names are GATE's proposal; what is not negotiable is each one's contract: who triggers it, what it takes in, what it delivers, and where it stops to ask.



Thick arrows come out of a person: those are the **five moments of human invocation**. Thin ones are chained by the framework. Dotted ones feed learning. Diamonds stop the chain until someone decides.

## Each skill's contract

| Skill             | Stage | Triggered by                         | Delivers                                                    | Writes outside, on approval |
|-------------------|-------|--------------------------------------|-------------------------------------------------------------|-----------------------------|
| /init             | 0     | Person                               | Dossier opened and classified by type of work               | Ticket                      |
| /define           | 1     | /init                                | Problem, context, data classification                       | Ticket                      |
| /specify          | 2     | /define                              | Acceptance and compliance criteria                          | Ticket                      |
| /refine           | 3     | /specify                             | Components, risks with owners, test and observability plans | Subtasks                    |
| /design           | 4     | /refine                              | Contrasted alternatives and a recorded decision             | Decision record             |
| /build            | 5     | Person                               | The change, implemented, instrumented and tested            | Repositories                |
| /self-review      | 6     | /build                               | Private findings, change proposal, QA script                | Change and script           |
| /review           | 7     | Person, several times                | Thread answers and its own findings                         | Comments, one by one        |
| /verify           | 8     | Person, with the /self-review script | Result per criterion                                        | Nothing                     |
| /deploy           | 9     | Person                               | Post-deployment check against the criteria                  | Production                  |
| /operate          | 10    | Alert or person                      | Diagnosis grounded in sources                               | Alerts and incidents        |
| /postmortem       | 10    | Incident closure                     | Timeline and root cause                                     | Document                    |
| /validate-outcome | 11    | /deploy , after the stage 2 window   | Outcome against the original problem                        | Ticket                      |
| /harvest          | 12    | /review and /postmortem              | Candidate rules and three lists                             | Catalog, rule by rule       |

Every skill declares explicitly **what it does not do**. An assistant with no written limits invents them on the fly, and invents them differently every time.

## The five moments of human invocation

People start the chain at five points, and only five: when the need is born ( `/init` ), when someone sits down to build ( `/build` ), when someone reviews ( `/review` ), when someone tests ( `/verify` ) and when someone deploys ( `/deploy` ). The framework chains everything else.

That matters because **mechanisms that depend on a person remembering to invoke them do not happen**. The classic symptom is an empty standards catalog next to changes carrying hundreds of discussion comments: nobody remembered to harvest.

## Cross-cutting skills

Two capabilities belong to no single stage and several stages consult them:

| Skill                        | What for                                                 | Consulted by                                                                                   |
|------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <code>/catalog</code>        | Fetch the rules in force that apply by scope and project | <code>/refine</code> , <code>/design</code> , <code>/self-review</code> , <code>/review</code> |
| <code>/false-positive</code> | Record a finding the team rejected, and the reason       | <code>/self-review</code> , <code>/review</code>                                               |

## 6. The actors

### The line between the AI and the people

The whole framework rests on a split that admits no exceptions.

#### The AI produces and verifies. The person judges and authorises.

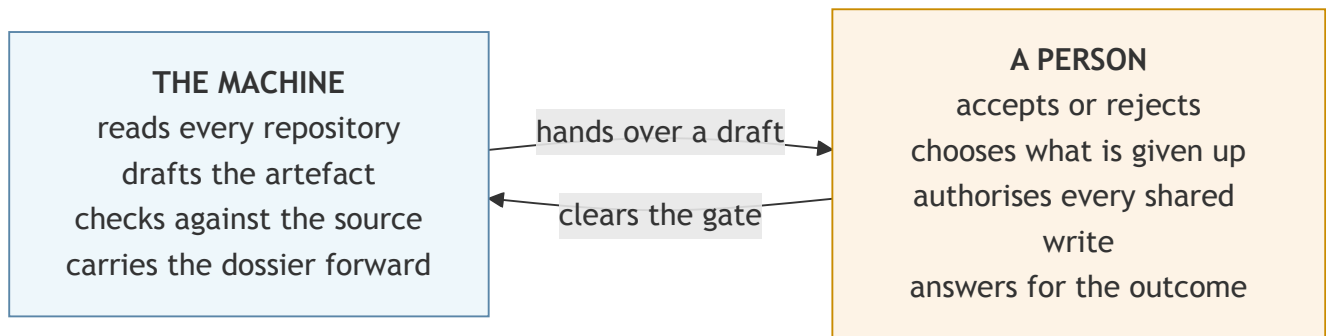
Four things the AI always does, at every stage:

1. Read and correlate whatever is needed, without tiring or skipping parts.
2. Draft every artefact, so nobody starts from a blank page.
3. Check against the source rather than against its memory, and say when it could not check.
4. Chain the next stage carrying the dossier, so context is never retyped.

Four things a person always does, and the AI never:

1. **Accept or reject at a gate.**
2. **Choose between alternatives when something has to be given up.** A trade-off between cost, time and scope is a business decision, not a calculation.
3. **Authorise any write into a shared system.**
4. **Answer for the outcome.** Responsibility is not delegated to a tool, and an organisation that tries finds out at the first audit.

When one of the four on the right moves to the left, the framework stops applying, however unchanged everything else looks.



The loop between those two boxes is the framework. The machine never crosses into the second one, and a person never has to do the first one by hand.

## The eight roles

Roles, not job titles. One person can hold several.

| Role                      | What they clear                     | Profile                                                                                     |
|---------------------------|-------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Requester</b>          | Nothing. They raise the need        | Knows the problem first hand. Needs no technical profile                                    |
| <b>Product owner</b>      | Gates 1, 2 and 11                   | Decides what gets built and what does not. Can take being told their hypothesis failed      |
| <b>Engineering lead</b>   | Gates 3 and 4                       | Architecture judgement and memory of the system. Able to reject a well-written proposal     |
| <b>Builder</b>            | Gate 6, over their own work         | Implements. Can read what the AI proposes and spot when it is wrong                         |
| <b>Reviewer</b>           | Gate 7                              | Someone other than the builder. Exercises judgement rather than re-reading what the AI said |
| <b>Verifier</b>           | Gate 8                              | Tests against the criteria. Can be another builder on the team                              |
| <b>Operations owner</b>   | Gate 9 and any action on production | Keeps the service up. Decides on rollback                                                   |
| <b>Standard custodian</b> | Gate 12                             | Approves rules. For hard rules, the whole team                                              |

## The three incompatibilities

One name can hold several roles, except in three cases. These are what keep judgement independent when the team is small.

**Whoever builds does not review that same change.** This is the separation that stops the assistant from ending up as the only reviewer.

**Whoever proposes a design decision does not accept it.** A single alternative presented without contrast, approved by the person who wrote it, is a decision nobody took.

**Whoever builds does not validate the business outcome.** Nobody judges their own work well.

### Who takes part in each stage

| Stage             | The AI does                                                   | The person does                                      | Who                          |
|-------------------|---------------------------------------------------------------|------------------------------------------------------|------------------------------|
| 0 · Entry         | Classifies the work, opens the dossier                        | Raises the need                                      | Requester                    |
| 1 · Definition    | Drafts the problem, flags ambiguity, classifies data          | Accepts the statement                                | Product                      |
| 2 · Requirements  | Proposes criteria, demands numbers on the non-functional ones | Accepts the criteria                                 | Product and engineering      |
| 3 · Refinement    | Maps components by reading the code, raises risks             | Assigns an owner to each risk, accepts the breakdown | Engineering                  |
| 4 · Design        | Puts forward contrasted alternatives and their consequences   | Picks one and answers for it                         | Engineering, not the builder |
| 5 · Development   | Implements, instruments, writes tests                         | Directs and corrects                                 | Builder                      |
| 6 · Pre-review    | Reviews in private, drafts the proposal and QA script         | Approves each publication                            | Builder                      |
| 7 · Change review | Answers threads, raises findings, harvests                    | Confirms each finding before it is published         | Reviewer                     |
| 8 · Verification  | Drafts the script                                             | Runs it and records results                          | Verifier                     |
| 9 · Deployment    | Compares against criteria, proposes rollback                  | Authorises every action on production                | Operations                   |
| 10 · Operations   | Diagnoses with evidence, drafts the postmortem                | Decides what gets touched and what gets alerted      | Operations                   |
| 11 · Validation   | Measures the outcome against the original problem             | Decides to close, iterate or roll back               | Product                      |
| 12 · Learning     | Drafts the candidate rule with its origin                     | Confirms it is a team decision                       | Custodian                    |

### How many people

Thirteen stages and eight roles do not mean thirteen people. Roles concentrate, and the three incompatibilities are what set the floor.

**Viable minimum: three people.** One in product, and two engineers who alternate: when one builds, the other reviews, and the other way round. That satisfies all three incompatibilities. Either engineer can carry operations.

**Recommended: four or five.** Add someone dedicated to operations, the role that survives part-time worst because incidents do not wait. And separate the requester from product, so the person asking is not the person approving.

**Practical ceiling: seven.** Above that the gates start queuing and the cycle stalls waiting for approvals. When a product needs more people, split it into two cycles with separate dossiers rather than fattening one.

A traditional cycle covering the same ground usually spreads these eight roles across eight or ten people, with a context handover at every step. The reduction does not come from anyone working more: it comes from context travelling in the dossier instead of being rebuilt in every handover meeting.

### **A warning about small teams**

Fewer people means fewer independent viewpoints. A small team with strong assistance converges fast, and can converge with great confidence on the wrong answer, because the assistant writes the wrong thing as fluently as the right one.

The three incompatibilities exist for exactly that. A team that cuts headcount and also relaxes them is not applying GATE: it is automating its own bias, faster than before.

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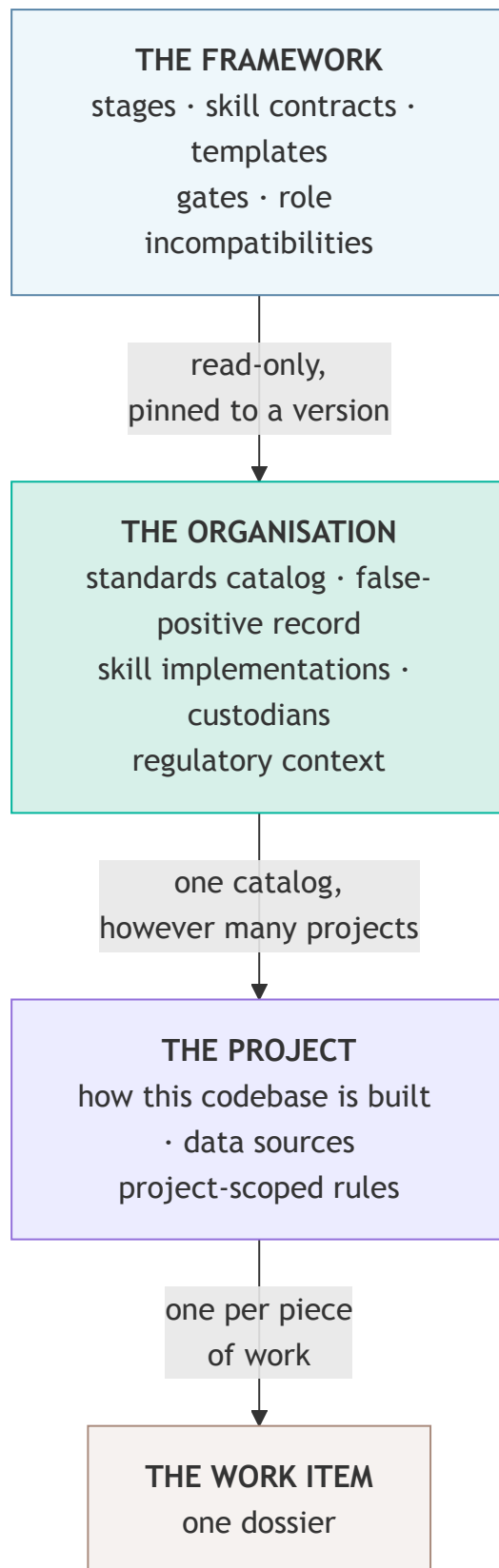
## **7. Scopes: one framework, many contexts**

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An organisation rarely has one product. It has products, services, internal tools and the odd isolated system that shares nothing with the rest. And a consultancy has several organisations at once.

GATE is the same door for all of them. What changes is not the cycle: it is **where each thing lives**.

## The four levels



| Level        | What lives there                                            | How many                   |
|--------------|-------------------------------------------------------------|----------------------------|
| Framework    | The method: stages, contracts, templates, gates             | One, for everybody         |
| Organisation | Catalog, false positives, skill implementations, custodians | One per organisation       |
| Project      | How that codebase is built, its data sources, its own rules | One per product or service |
| Work item    | The dossier                                                 | One per piece of work      |

**One catalog per organisation, not one per project.** This is the decision that matters most. With a catalog per project, a rule learned in one never reaches another, and the compounding that pays for the whole framework disappears. You would have twenty silos learning the same lesson twenty times.

Scoping is what keeps that from becoming noise: a rule carries `projects`, so it applies where it belongs and stays out of stacks it has nothing to say about.

### Isolated contexts

A service that shares nothing with the rest still enters through `/init` and runs the same thirteen stages. It is simply a project whose rules never leave project scope.

That is not a lesser case. Being able to run the cycle on something isolated, with no catalog worth speaking of and no other project to learn from, is what makes GATE usable on day one, by one team, before any of this topology exists.

### What travels, and what never does

**Only a rule travels upward.** Everything else stays where it was created.

A rule is born scoped to the project where it was harvested. It reaches organisation scope when there is evidence it applied in a second project, and that promotion is a human decision recorded in the catalog like any other.

Four things **never** cross an organisation boundary, and this is not a preference:

- **Dossiers.** They carry the client's problem, their data classification and their incidents.
- **Data sources.** Logs, metrics and audit trails belong to whoever owns the system.
- **Credentials.** Personal, per person, per organisation.
- **The project definition.** It describes a codebase that belongs to someone.

For a consultancy this is the line that keeps one client's material out of another's. It is worth stating in the contract, not just in a document.

### Between organisations

The framework is shared: it is public and identical for everyone. The catalogs are not.

A rule does not cross from one organisation's catalog to another automatically, ever. Risk appetite, regulation and legal exposure differ, and a hard rule that is right for one can be wrong or simply inapplicable for the next.

What does travel is a person carrying a lesson. The mechanism is to **propose** it in the other organisation's catalog as a candidate, with the origin pointing at where it was learned. It then goes through that organisation's own approval like any other candidate.

## Governance once there are many projects

Three things change, and none of them is technical.

**The custodian stops being a person.** With one project, whoever knows the codebase can hold the standard. With twenty it needs a small group, and a rule for what happens when they disagree.

**"The whole team approves hard rules" needs a definition.** The framework says a hard rule requires the team's approval. Once there is no single team, you have to say which one: the affected projects' leads, a standards group, or whoever your organisation already trusts with cross-cutting decisions.

**Compliance becomes a question you have to be able to answer.** Which projects satisfy rule RD-0007. With one project you know from memory. With twenty you need to report it, and the framework does not provide that today: the catalog says what the rule is and the dossier says what was applied to one change, but nothing aggregates the two across projects.

That gap is real and it is named here rather than glossed over. The cheapest honest answer for now is that each project's pre-review already reads the catalog, so non-compliance surfaces on the next change to that project rather than on a report.

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## 8. The stages

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### Stage 1 · Definition

Turns a need in business language into a statement an engineering team can assess. Product runs it.

The assistant drafts the statement separating the observed problem from the solution someone already imagined: most requests arrive with a solution inside and without the problem that motivated it. It flags ambiguity instead of resolving it, returning both possible readings. It classifies the data involved — personal, health, financial, regulated — and that classification travels with the dossier to production, setting the depth of every later stage. And it looks for precedent: earlier tickets, decisions already taken, related incidents, catalog rules that apply.

**Gate.** Product accepts. Nothing moves on without a data classification.

### Stage 2 · Requirements

Settles what done means, in terms that can be checked.

The assistant proposes acceptance criteria that can be verified, and demands numbers on the non-functional ones: response time, expected volume, required availability, tolerable recovery window. Without a number they cannot be verified in stage 8 or watched in stage 10. It derives the compliance requirements

that follow from the data classification: retention, audit, encryption, minimisation, data subject rights. And it points out which criteria will need instrumentation, so stage 3 accounts for it.

**Gate.** A non-functional criterion without a number goes back.

### Stage 3 • Refinement

Turns requirements into executable work, with the risks on the table before any code.

The assistant breaks the work into tasks with verifiable outputs, and maps the affected components **by reading the actual code**: a map drawn from memory or from an old diagram is wrong precisely about the systems that changed the most. It raises security and compliance risks grounded in the data classification and the hard rules in force. It detects when a requirement collides with a rule in force and raises it as a product decision. It proposes the test plan and the **observability plan**: whatever is not instrumented here does not exist in stage 10.

**Gate.** A risk without an owner blocks the move to development.

### Stage 4 • Design

Settles the architecture decision before any code, and puts it on the record.

This is where assistance pays off most and carries most risk: the assistant proposes a solution as fluently whether it is well founded or not. That is why the decision is recorded, not just the result.

It puts forward **at least two workable alternatives** with their consequences, rather than presenting one as if it were the only option. It names what each alternative closes off for the future, which usually weighs more than its cost today. It checks each one against the rules in force and against the patterns the project already holds. And it writes a short decision record: what was decided, what was ruled out and why, and what would have to change to revisit it.

**Gate.** Someone who did not draft the proposal accepts the decision. A single alternative presented without contrast goes back.

This record is one of the best sources for stage 12: a design decision already arrives argued and contrasted, which is exactly what a rule usually lacks.

### Stage 5 • Development

Implement, leaving the trail the later stages need. The assistant consults the catalog before proposing a pattern, instruments what the plan called for in the same change rather than afterwards, and writes the tests alongside the code.

### Stage 6 • Pre-review

First of the two reviews. It is **private**: its output goes to the author. It exists so the change reaches human review already cleaned up.

It identifies the ticket from the branch and pulls its definition without editing it. It determines every repository the change touches and checks none was left half done. It runs linters and static analysis. It checks every claim about conventions against the actual code. It runs a full hard-rule pass if the change

touches authentication, authorisation, tenant isolation or sensitive data. It cross-checks the artefacts that travel together: migrations with their changelog entry, instrumentation with the plan, documentation with the code. It compares **the whole branch against the base**, not just the latest commits, against the criteria from stage 2. And it drafts the change proposal and the functional verification script.

**Gate.** The author approves every output into a shared system, one at a time.

## Stage 7 · Change review

Someone other than the author validates the change, with support. **It runs several times** while the review lasts: a single pass when the change opens arrives before the decisions exist.

It reads the threads across every change in the ticket and splits them into open ones, which need answering, and **resolved** ones, which are the input for stage 12. It checks every suggestion from another reviewer against the actual code before backing it or pushing back. It raises findings with severity, file and line, and with the rule id when the finding comes from the catalog: without that id nobody can go and argue with the rule, and arguing with it is how the standard improves. It consults the false-positive record before raising something already rejected.

**Gate.** Every finding is presented with an explicit counterfactual question — whether it really applies here or is a false positive — and is published only on confirmation.

**Why two reviews and not one.** The pre-review is private and exists so the author can fix things at no social cost. The change review is public and exists so another person exercises judgement. Merging them turns the assistant into the reviewer, which is exactly what this framework avoids.

## Stage 8 · Functional verification

Check against the criteria from stage 2, on the running system. The assistant drafts the script; it neither runs it nor signs it off.

A good script does six things: one checkbox per **verifiable result** rather than per case, so a failure points at what failed; one block per affected component, naming what to look at on each screen; the precondition written as a check the tester can perform, with what to do if it is not met; steps executable with the tools they already have; known false alarms stated up front; and what to do with whatever cannot be tested in that environment, which is to leave it unchecked and note it, never mark it failed.

**Gate.** A stage 2 criterion left unverified blocks deployment.

## Stage 9 · Deployment and post-deployment check

Put the change in production and check with data that it behaves the way the criteria said.

The assistant gathers the manual steps the deployment requires instead of letting them surface during the window. It compares observed behaviour against the non-functional criteria from stage 2 using the data sources from stage 10: that comparison is why those criteria had to carry a number. It watches the window afterwards for deviations from the previous baseline, not from an invented threshold. On detecting a deviation it presents the evidence and **proposes** a rollback; the decision is human.

**Precondition.** A rollback procedure that has been exercised, not merely documented.

## Stage 10 · Operations, SRE and continuity

Keep the service up, detect before the user does, and recover inside the committed window.

### The data sources

An assistant with no access to operational data offers opinions. With access, it diagnoses.

| Source                     | What it answers                              | Common mistake                                                             |
|----------------------------|----------------------------------------------|----------------------------------------------------------------------------|
| Application logs           | What the system did and with what error      | Taking an intermediary's log as if it were the origin's                    |
| Metrics                    | How much and how fast, over time             | Watching averages and losing the high percentiles, where the problem lives |
| Distributed traces         | Where a request went and where the time went | Not propagating the correlation id and losing the chain                    |
| Infrastructure logs        | What the platform did underneath the service | Blaming the application for node rotation or network                       |
| Audit trail                | Who did what, when, over which data          | Mixing it with application logs and losing its evidential value            |
| External dependency status | Whether the problem is ours or inherited     | Diagnosing inward a failure that belonged to a third party                 |

Three conditions make them useful. A **correlation id propagated end to end**, without which each source tells a different story. **One source of truth per question**: when two dashboards answer the same question with different numbers, the team stops believing both. And **declared, sufficient retention**, which follows from the compliance requirements of stage 2, because an incident investigated three days later needs data from three days ago.

### What the assistance does

It correlates the sources from a symptom and separates cause from noise, grounding every claim in the source that can prove it. It compares current behaviour against the baseline before the last change and links it to the ticket that introduced it. It proposes what deserves an alert and, above all, what does not: if nobody will do anything differently when it fires, it is a dashboard number. It drafts the postmortem with the timeline rebuilt from the sources, not from the memory of whoever was on call. And it prepares the recovery drill script, comparing the real result against the committed window.

### Securing operations

Five conditions. None is exotic and almost none is met in full.

**A rollback that has been exercised.** A procedure never run is a hypothesis.

**Recovery windows with a number and a drill.** How long you tolerate being down and how much data you tolerate losing, declared in stage 2 and measured in a real drill.

**External dependencies with defined degradation.** What the system does when a third party does not answer. Without that definition, the default answer is to pass the failure to the user.

**Blast radius isolation.** That one component failing does not drag the rest. Proven by causing it in a controlled environment.

**Audited emergency access.** With a trail and an expiry. Permanent emergency access is normal access under a misleading name.

## **Stage 11 · Outcome validation**

Go back and ask whether the problem from stage 1 was solved.

This is the stage almost no cycle has, and its absence is why organisations measure what they shipped rather than what worked. Stages 8 and 9 check that the system does what the criteria said. Neither checks that this solved anything.

It runs once the change has been in production long enough to have data, and that window is set in stage 2 alongside the criteria.

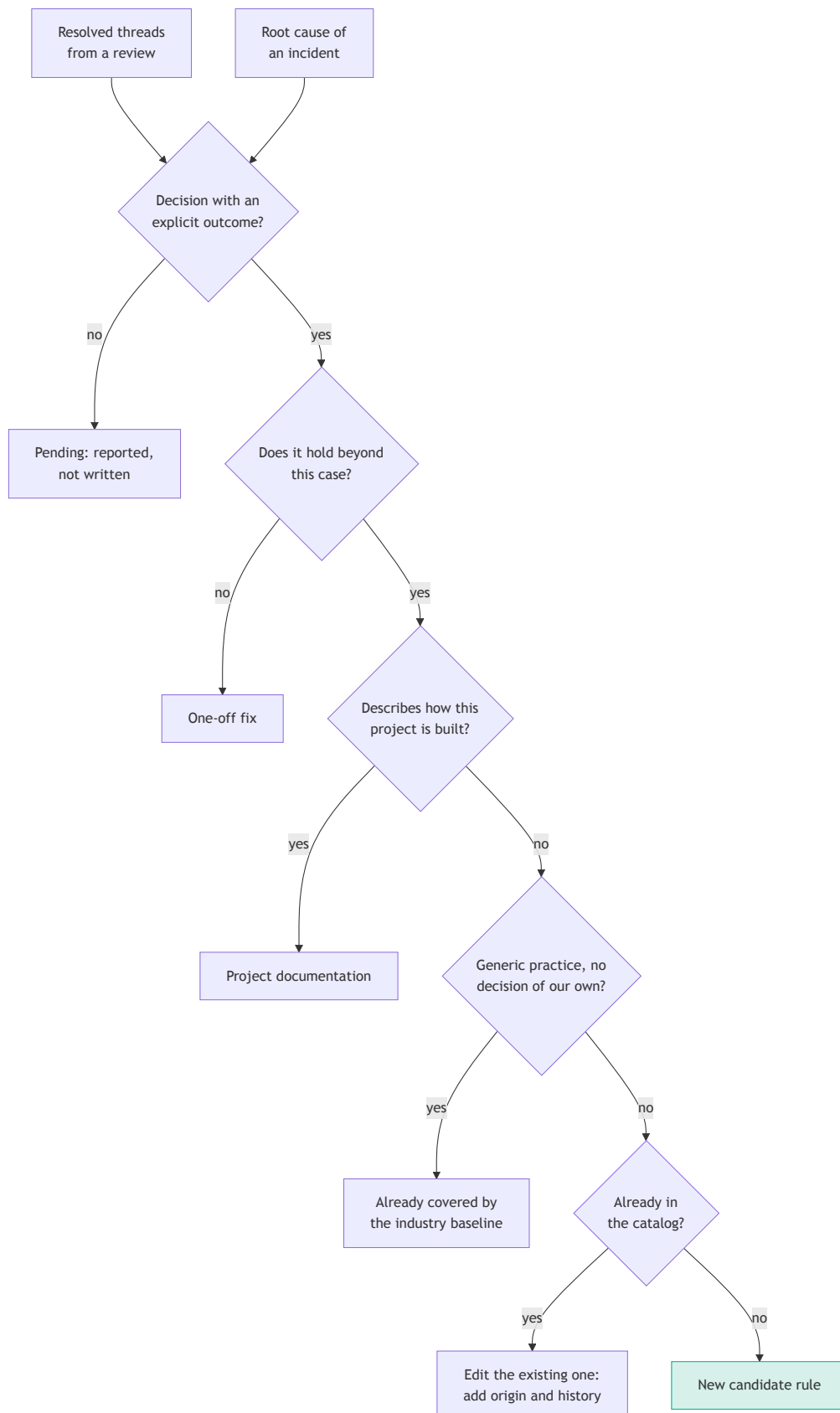
The assistant compares observed behaviour in production against the problem stated in stage 1, using the data sources from stage 10 and whatever business indicators apply. It distinguishes three outcomes and names them without softening: the problem was solved, it was half solved and here is what was left out, or it was not solved and the stage 1 hypothesis was wrong.

**Gate.** Product accepts the outcome and decides: close, iterate, or roll the feature back.

This is the framework's **second learning loop**. The stage 12 loop improves how things get built; this one improves what gets decided. A product hypothesis that failed is as valuable as a root cause, and is lost just as easily.

## **Stage 12 · Learning**

Make what was learned outlive the conversation or the shift where it appeared. **Two sources, one destination.**



An incident is as good a source of rules as a discussion, and usually yields better ones: nobody argues with a root cause that already cost an outage.

**Gate.** Every rule is presented with a counterfactual question — whether it is a team decision or the assistant's general opinion with a link stapled to it.

**Output.** Candidate rules and a report with three lists, delivered every time even when the first is empty: harvested, discarded with the reason, and pending decisions.

A rule is born **with no blocking power**. Acquiring it is an explicit human decision, and for security rules, the whole team's. That stops the assistant from imposing standards nobody agreed to.

### **Optional stage · Retirement**

Switching off a system or a feature carries obligations almost no cycle covers, and it is where compliance risk piles up quietly.

What has to be settled: which data must be kept and for how long, which must be deleted and with what evidence, which integrations depend on what is being switched off, and who owns the historical archive afterwards. The data classification from stage 1 says what applies.

Adopt it when the organisation actually retires systems. One that never switches anything off does not need the stage, and usually has the problem of never switching anything off.

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## **9. Effect on the organisation**

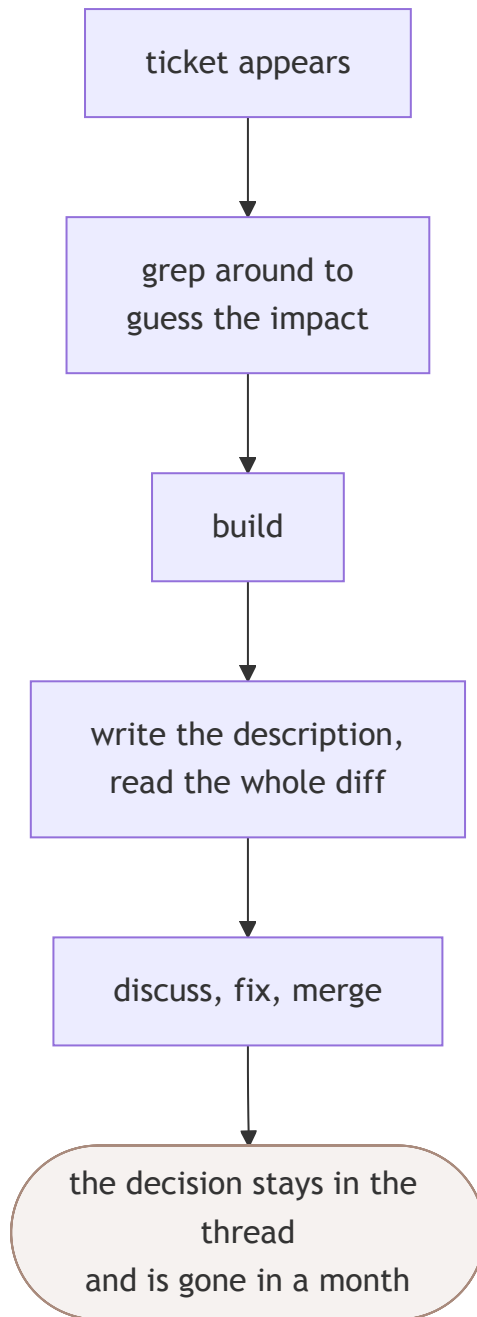
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### **Where the time goes today**

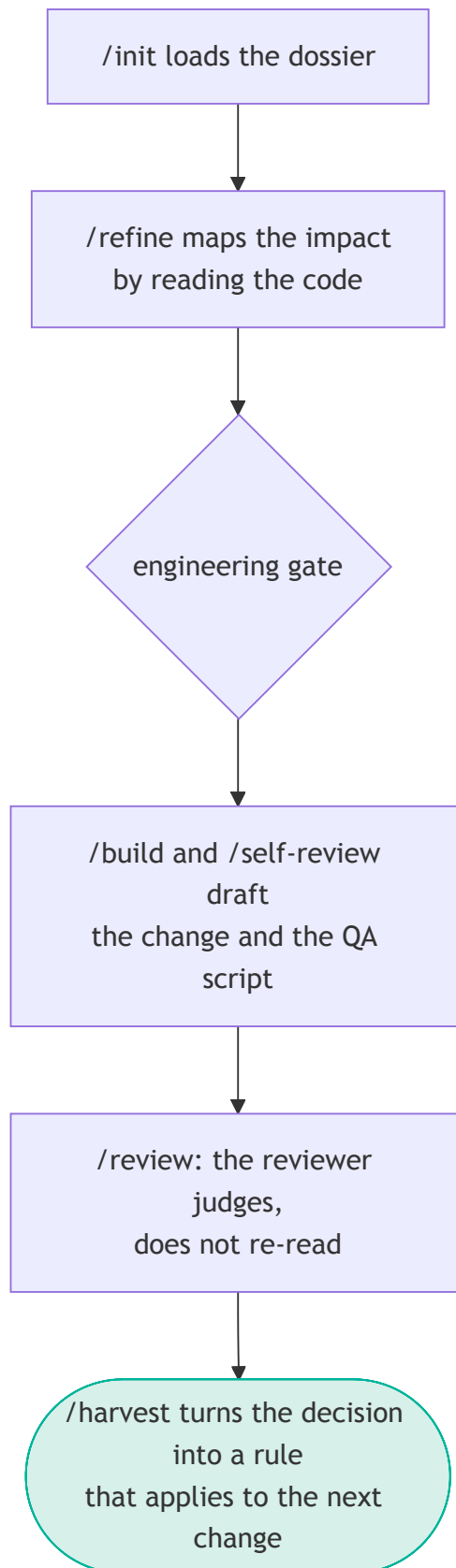
In a traditional cycle with no or limited AI support, much of the effort is not spent deciding. It is spent producing and carrying artefacts: drafting the ticket, explaining the context again at refinement, reading the whole change to review it, writing the test script, rebuilding an incident timeline by hand, and drafting the postmortem nobody will reread.

That work is necessary and it is mechanical. It is exactly where assistance pays.

### **One change, the usual way**



**The same change, with GATE**



The mechanical steps on the first line — grepping for impact, writing the description, reading the whole diff — are the ones that collapse. What does not collapse is the judgement: the gate in the middle and the reviewer near the end.

And the last step on the second line has no equivalent on the first. That is where the compounding comes from: the same discussion does not happen twice.

### What changes with GATE

Mechanical work collapses and **the bottleneck moves to judgement**. What used to take an afternoon of drafting becomes a review of something already drafted. The gates stop being paperwork and become the work.

That has three consequences worth stating plainly.

**The same team covers more ground.** Not because anyone works more, but because they stop producing intermediate artefacts by hand.

**Roles consolidate.** The line between whoever refines and whoever builds, or between whoever operates and whoever diagnoses, blurs once context travels in the dossier instead of in someone's head. A smaller team can sustain a product that used to demand several separate roles.

**The cost of a bad decision rises.** When the assistant produces in minutes what used to take days, a wrong call propagates just as fast. That is why the gates are mandatory and why the two reviews are kept apart.

### The warning

Cutting headcount is a possible consequence, not a goal of the framework, and it carries a concrete risk: fewer people means fewer independent viewpoints.

A small team with strong assistance can converge quickly on the wrong answer, and converge confidently, because the assistant writes the wrong thing as fluently as the right one.

The countermeasure is in the design: that pre-review and change review are never the same person, that the product and engineering gates are held by someone who did not write what is being approved, and that no rule reaches blocking status without the team's approval. A team that cuts people and also relaxes those three points is not applying GATE: it is automating its own bias.

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## 10. What improves, and how to check it

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Without indicators, this is a well-written opinion.

**Productivity.** Time between a change being ready and entering review. Time to diagnose an incident.

**Quality.** The share of findings that surface in pre-review versus those that reach human review. If the second number does not fall over time, the cycle is not learning.

**Security.** How many security findings are caught before any code is written.

**Compliance.** That for any change in production you can answer which standard was applied, with which rule, at which version, approved by whom. A versioned catalog answers that; a conversation with an assistant does not.

**Continuity.** Recovery drill results against the committed window. The share of incidents detected by your own instrumentation before a user reports them.

**Learning.** How many rules entered the catalog per quarter, and how many repeated discussions stopped happening. A catalog that does not grow means stage 12 is not running, whatever the other eleven say.

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## 11. Preconditions

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- Personal credentials per person, never shared. A team token destroys the traceability of who approved what.
  - A local copy of every repository the cycle touches, including the standards one.
  - A catalog readable by machine, versioned, with a format validator.
  - A low-friction place to record false positives, outside the rule approval process. If recording a false positive cost an approval, nobody would record one.
  - A branch convention that links to the ticket, so traceability resolves without asking.
  - A correlation id propagated end to end, and a declared retention period.
  - A place for the dossier to live, readable by people and by machine.
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## 12. Adoption

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In this order. Each step works without the ones after it, and none requires the previous one to be perfect.

1. **The standards catalog, empty.** With a rule format, a validator and a generated catalog. Empty on purpose: a set copied from an industry guide describes whoever copied it.
2. **The pre-review.** It returns the most for what it costs and forces nobody else to change how they work.
3. **The change review, with harvesting inside it.** Both together. An assisted review that does not harvest leaves the cycle open.
4. **The functional verification script.** Cheap, and it draws a clear line between review and QA.
5. **Refinement and design.** Once the catalog holds rules worth consulting before settling an architecture.
6. **Operations and SRE.** Requires the data sources to exist and be trustworthy, usually the longest work of all.
7. **Stage 0, definition and requirements.** They depend on product picking up the habit, and that cannot be decreed.
8. **Outcome validation.** Last, because it needs trustworthy business data and the willingness to record that a hypothesis failed.

The framework pays from step 2. There is no need to wait for the whole thing.

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## 13. Anti-patterns

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**The assistant that acts on its own.** Publishes, opens changes or touches environments without approval. One visible mistake is enough for the team to lose trust and go back to doing everything by hand.

**The chain with no gates.** A cycle that runs end to end without human intervention produces code with no owner, and nobody can answer why it was done that way.

**The catalog filled in one sitting.** Fifty rules copied from a generic guide. Nobody recognises them as their own and the team learns to ignore the findings.

**Manual harvesting.** A command someone should run when they remember. It does not happen.

**The single-pass review.** Decisions appear when threads get resolved, and by then nobody is looking.

**The silent discard.** Rejecting a finding without recording the reason.

**The rule with no origin.** A norm nobody can trace gets re-argued every time it appears.

**Observability as a later task.** The first incident is diagnosed blind.

**The alert nobody acts on.** It trains the team to ignore the whole channel.

**The non-functional criterion with no number.** It cannot be verified in QA or watched in production, so in practice nothing was required.

**The assistant opining on production without data.** A diagnosis with no sources is a well-written guess, which is worse than none.

**The abandoned dossier.** If one stage stops writing to it, the next one asks again what was already answered, and the chain breaks without anyone noticing.

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## About the name

**GATE** — Governed, Assisted, Traceable Engineering.

Four words, each carrying its own weight. *Governed*: a person clears every control point. *Assisted*: the machine does the reading, the correlating and the drafting. *Traceable*: the dossier answers what was asked, what standard was applied and who approved it. *Engineering*: the scope is building and running software, not organising the team.

The word itself is the thesis. While most of the field is selling autonomous agents, this framework sells the places where the work stops and a person decides.

Every gate in it exists because something, somewhere, once went through without one.